Debug With Shubham Notes

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tcs NQT 2025

DEBUG WITH SHUBHAM
TECH CHANNEL

2-3 Days Interview Plan & Que & Ans

Ninja Role 2025

DAY 1: Core Technical (Coding)

🕒 Time Needed: 10 hours

🎯 Focus: Logic building, basic DSA

✅ Patterns(star, number, pyramid)

✅ Array (frequency, duplicates, reverse)

✅ String (palindrome, anagram, character count)

Practice Sites:

LeetCode
GeeksforGeeks

➡️ Target: Solve 10-15 questions that match past TCS patterns.

Day 2- SQL + DMBS + Core Sub

✅ SELECT, WHERE, GROUP BY, HAVING

✅ JOINs – especially INNER JOIN, LEFT JOIN

🔷 CS Basics:

- ✅ OOP Concepts (Encapsulation, Inheritance, Polymorphism, abstraction)
- ✅ DBMS (Normalization, Keys)
- ✅ OS basics (Deadlock, Paging)
- ✅ Networking (TCP/IP, HTTP vs HTTPS)

Day 3- HR & Managerial Questions + Project

Tell me about yourself

Why TCS / Ninja role?

Strengths and Weaknesses

Describe a project (pick 1 from your resume)

What if you're rejected?

➡️ Tip: Practice in front of a mirror or record yourself

Project:

Goal/Purpose

Technologies Used

Your Role

Challenges & Solutions

Output / Result

Coding Question

1-Reverse a String
Input: "hello"
Output: "olleh"

2-Check if a number is Prime
Input: 17
Output: Prime

3-Find the Factorial of a Number
Input: 5
Output: 120

4-Fibonacci Series up to N terms
Input: 5
Output: 0 1 1 2 3

5- Palindrome String Check
Input: "madam"
Output: Palindrome

Find the Largest Element in an Array
Input: [3, 7, 2, 5]
Output: 7

Sum of Digits of a Number
Input: 123
Output: 6

Print Star Pattern (Right Triangle)
Input: 4
Output:

```
*
* *
* * *
* * * *
```

Check Armstrong Number
Input: 153
Output: Armstrong
$(1^3 + 5^3 + 3^3 = 153)$

Count Frequency of Each Character
Input: "aabbc"
Output: a:2, b:2, c:1

BONUS (Asked in TCS NQT Interview Frequently)

- 🔶 Find the Second Largest Number in an Array
- 🔶 Remove Duplicates from an Array
- 🔶 Find GCD of Two Numbers
- 🔶 Find Missing Number in 1-N Sequence
- 🔶 Swap Two Numbers Without Temp Variable

SQL

1-Select All Columns from a Table

```
SELECT * FROM Employees;
```

2-Find the Second Highest Salary

```
SELECT MAX(Salary)
FROM Employees
WHERE Salary < (SELECT MAX(Salary) FROM Employees);
```

3-Find Employees from a Specific Department

```
SELECT *
FROM Employees
WHERE Department = 'IT';
```

4-Count the Number of Employees in Each Department

```
SELECT Department, COUNT(*)
FROM Employees
GROUP BY Department;
```

5-Display Total Salary by Department

```
SELECT Department, SUM(Salary)
FROM Employees
GROUP BY Department;
```

6-JOIN Two Tables (INNER JOIN)

```
SELECT e.Name, d.DeptName
FROM Employees e
INNER JOIN Departments d ON e.DeptID = d.DeptID;
```

7-Find Employees Who Don't Have a Manager

```
SELECT Name
FROM Employees
WHERE ManagerID IS NULL;
```

8-Find Duplicate Employee Names

```
SELECT Name, COUNT(*)
FROM Employees
GROUP BY Name
HAVING COUNT(*) > 1;
```

9-Find Employees Hired in the Last 6 Months

```
SELECT *
FROM Employees
WHERE HireDate >= DATEADD(MONTH, -6, GETDATE());
```

10-Subquery to Get Employee with Max Salary

```
SELECT *
FROM Employees
WHERE Salary = (SELECT MAX(Salary) FROM Employees);
```

BONUS Quick Topics:

- DISTINCT, ORDER BY, LIMIT
- IN, BETWEEN, LIKE, IS NULL
- Difference between WHERE and HAVING
- Types of joins (LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN)

DBMS

1-What is DBMS?

A Database Management System is software that manages databases and allows CRUD operations (Create, Read, Update, Delete).

2-What is the difference between DBMS and RDBMS?

DBMS	RDBMS
No relationships	Maintains relations (tables)
No keys required	Uses Primary and Foreign Keys
Less secure	More secure

3-What are Primary and Foreign Keys?

Primary Key uniquely identifies each record (e.g., EmpID)

Foreign Key links one table to another (e.g., DeptID in Employees referencing Departments)

4-What is Normalization?

Normalization is the process of organizing data to reduce redundancy and improve integrity.

1NF: No repeating groups

2NF: No partial dependency

3NF: No transitive dependency

5-What are the types of SQL commands?

DDL: CREATE, ALTER, DROP

DML: INSERT, UPDATE, DELETE

DCL: GRANT, REVOKE

TCL: COMMIT, ROLLBACK

DQL: SELECT

6-What is a Transaction?
A transaction is a single unit of work with ACID properties:

Atomicity
Consistency
Isolation
Durability

Example: Transferring money between bank accounts.

7-What is a View?
A view is a virtual table based on the result of a SQL query.

Difference Between WHERE and HAVING

WHERE	HAVING
Filters rows	Filters groups
Used before GROUP BY	Used after GROUP BY

8-What is Indexing?

Indexing improves search speed on large tables.

Like the index of a book — makes finding faster.

9- Difference: DELETE vs TRUNCATE vs DROP

DELETE: removes rows (can rollback)

TRUNCATE: removes all rows (no rollback)

DROP: deletes the entire table

🧠 Bonus Topics:

- Candidate Key vs Super Key
- E-R Model (Entity, Attributes, Relationships)
- Types of Relationships (1:1, 1:N, M:N)
- JOIN vs Subquery
- Denormalization